

Problem

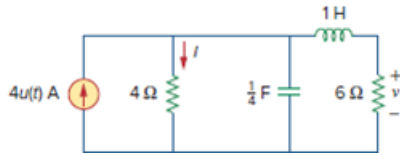
Refer to the circuit in Fig. 8.66. Determine:

(a) $i(0^+)$ and $v(0^+)$,

(b) $di(0^+)/dt$ and $dv(0^+)/dt$,

(c) $i(\infty)$ and $v(\infty)$.

Figure 8.66:



Step-by-step solution

Step 1 of 12

Refer to Figure 8.66 in the textbook.

(a)

For $t < 0$, the source current is, $4u(t) = 0$ A. So, the circuit is a source free circuit for $t < 0$.

In a source free circuit, all currents and voltages are zero.

The current through the inductor at $t < 0$ is,

$$i_L(0^-) = 0 \text{ A}$$

The voltage across the capacitor at $t < 0$ is,

$$v_C(0^-) = 0 \text{ V}$$

The voltage across the capacitor cannot change instantaneously.

$$\begin{aligned} v(0^+) &= v(0^-) \\ &= 0 \text{ V} \end{aligned}$$

Step 2 of 12

Draw the circuit diagram by replacing the inductor initial current and capacitor initial voltage.

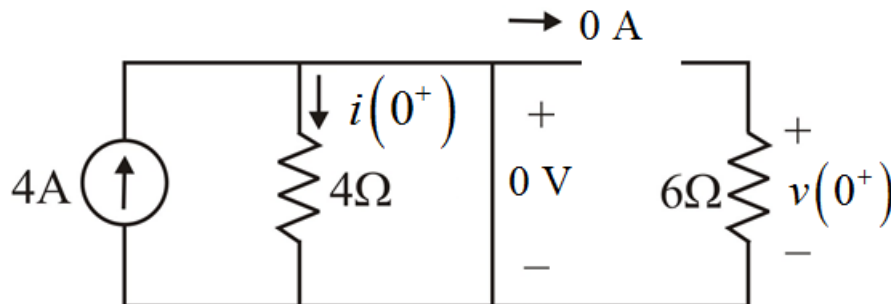


Figure 1

Step 3 of 12

Calculate the value of $i(0^+)$.

$$\begin{aligned} i(0^+) &= \frac{v_c(0^-)}{4} \\ &= \frac{0}{4} \\ &= 0 \text{ A} \end{aligned}$$

Calculate the value of $v(0^+)$.

$$\begin{aligned} v(0^+) &= i_L(0^-) \times 6 \\ &= 0 \times 6 \\ &= 0 \text{ V} \end{aligned}$$

Therefore, the value of current at $t = 0^+$ is $i(0^+) = 0 \text{ A}$ and the value of voltage at $t = 0^+$ is $v(0^+) = 0 \text{ V}$.

Step 4 of 12

(b)

Draw the circuit diagram for $t > 0$.

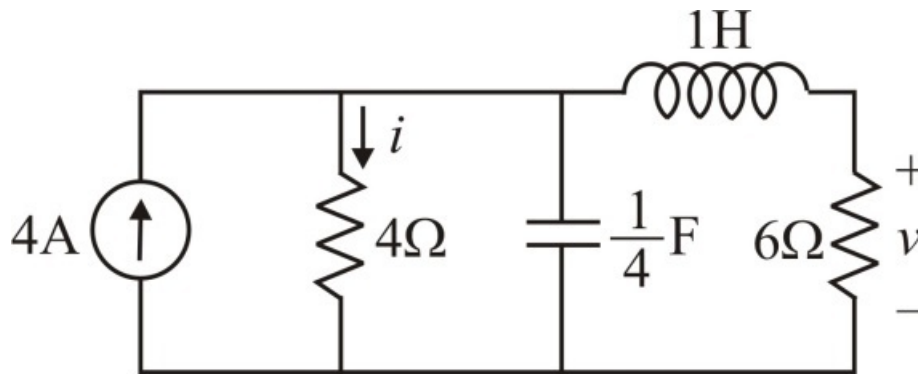


Figure 2

Step 5 of 12

Apply Kirchhoff's current law at top node.

$$\begin{aligned} -4 + i(t) + i_c(t) + i_L(t) &= 0 \\ -4 + i(t) + C \frac{dv_c(t)}{dt} + i_L(t) &= 0 \end{aligned}$$

At $t = 0^+$,

$$\begin{aligned} -4 + i(0^+) + \frac{1}{4} \frac{dv_c(0^+)}{dt} + i_L(0^+) &= 0 \\ -4 + 0 + \frac{1}{4} \frac{dv_c(0^+)}{dt} + 0 &= 0 \\ \frac{1}{4} \frac{dv_c(0^+)}{dt} &= 4 \\ \frac{dv_c(0^+)}{dt} &= 16 \text{ V/s} \end{aligned}$$

Step 6 of 12

From the Figure 2, it is clear that the voltage across 4Ω resistor is same as capacitor voltage.

$$v_C(t) = i(t) \times 4$$

$$i(t) = \frac{1}{4} v_C(t)$$

$$\frac{di(t)}{dt} = \frac{1}{4} \frac{dv_C(t)}{dt}$$

At $t = 0^+$,

$$\begin{aligned} \frac{di(0^+)}{dt} &= \frac{1}{4} \frac{dv_C(0^+)}{dt} \\ &= \frac{1}{4} \times 16 \\ &= 4 \text{ A/s} \end{aligned}$$

Thus, the value of $\frac{di(0^+)}{dt}$ is 4 A/s.

Step 7 of 12

Apply Kirchhoff's voltage law to right hand side loop of Figure 2.

$$-v_C(t) + v_L(t) + v(t) = 0$$

At $t = 0^+$,

$$-v_C(0^+) + v_L(0^+) + v(0^+) = 0$$

$$-0 + v_L(0^+) + 0 = 0$$

$$v_L(0^+) = 0$$

Step 8 of 12

Consider the expression for voltage across the inductor.

$$v_L(t) = L \frac{di_L(t)}{dt}$$

At $t = 0^+$,

$$v_L(0^+) = L \frac{di_L(0^+)}{dt}$$

$$0 = (1) \frac{di_L(0^+)}{dt}$$

$$\frac{di_L(0^+)}{dt} = 0$$

Step 9 of 12

From the Figure 2, it is clear that the voltage across 4Ω resistor is same as capacitor voltage.

$$v(t) = i_L(t) \times 6$$

$$\frac{dv(t)}{dt} = 6 \frac{di_L(t)}{dt}$$

At $t = 0^+$,

$$\frac{dv(0^+)}{dt} = 6 \frac{di_L(0^+)}{dt}$$

$$\frac{dv(0^+)}{dt} = (0)(6)$$

$$\frac{dv(0^+)}{dt} = 0 \text{ V/s}$$

Thus, the value of $\frac{dv(0^+)}{dt}$ is 0 V/s .

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(c)

For $t = \infty$, the capacitor acts as an open circuit and inductor acts as a short circuit.

Draw the modified circuit diagram.

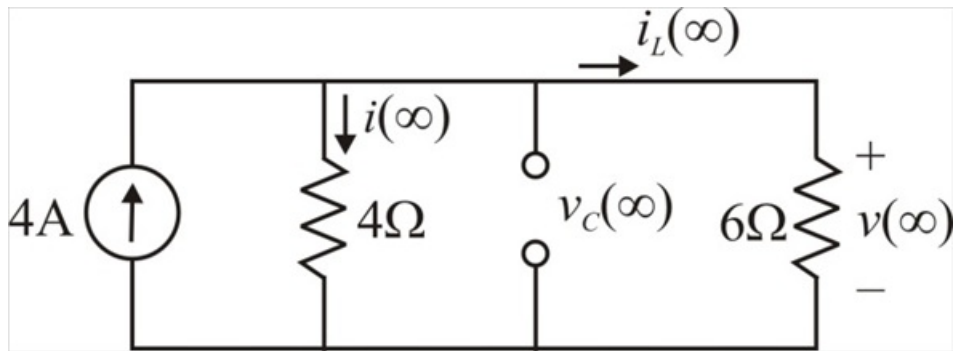


Figure 3

Step 11 of 12

Apply current division rule to calculate the final current, $i(\infty)$.

$$\begin{aligned} i(\infty) &= 4 \left(\frac{6}{4+6} \right) \\ &= \frac{24}{10} \\ &= 2.4 \text{ A} \end{aligned}$$

Therefore, the current through 4Ω resistor at $t = \infty$, $i(\infty)$ is 2.4 A .

Step 12 of 12

Calculate the value of $i_L(\infty)$.

$$\begin{aligned} i_L(\infty) &= 4 - i(\infty) \\ &= 4 - 2.4 \\ &= 1.6 \text{ A} \end{aligned}$$

Calculate the voltage across $6 \text{ k}\Omega$ resistor at $t = \infty$.

$$\begin{aligned} v(\infty) &= i_L(\infty)(6) \\ &= (1.6)(6) \\ &= 9.6 \text{ V} \end{aligned}$$

Therefore, the voltage across $6 \text{ k}\Omega$ resistor at $t = \infty$, $v(\infty)$ is 9.6 V.